

Girish Narayanswamy

ML/AI || Health Sensing || Signal Processing

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Girish is a final year PhD Student at the University of Washington **Ubiquitous Computing Lab**, where he is advised by Professor **Shwetak Patel**. Girish's research focuses on novel **Learning (ML/AI)** and **Time-Series Modeling** methodologies. He is particularly interested in applications which improve **health sensing** and expand **health access**. Note that *pink* denotes clickable linked-text.

Education

University of Washington Seattle

PhD Electrical & Computer Engineering

Advised by Shwetak Patel || UbiComp Lab

Sept. '20 - Present



University of Colorado Boulder

BS Electrical & Computer Engineering (Magna Cum Laude & Eng. Honors)

Minor Computer Science

Aug. '14 - May '19

GPA: 3.86

Selected Projects



Influenza Onset Prediction via Wearables (Preparing for Nature)

Oct. '23 - Present

- In collaboration with NIH/NIAID, deployment of commodity sensors/wearables to participants of NIH influenza challenge study.
- Study participants challenged with H10N7 influenza - early results validate the use of wearables in detecting the onset of influenza like illnesses.



Wearable Sensor Foundation Models (ICLR'25, NeurIPS'25)

June '24 - Present

- Development of **Large Sensor Models (LSM)** – wearable health foundation models.
- Featured at Google's public forum ([The Check Up](#)) and health.google/ai-models/
- **LSM-1 [8]**: First evidence of scaling in the wearable domain.
Blog: research.google/blog/scaling-wearable-foundation-models
- **LSM-2 [UR.6]**: Improving scaling by handling fragmented data.
Blog: research.google/blog/lsm-2-learning-from-incomplete-wearable-sensor-data
- **SensorLM [UR.5]**: Language + sensor pairwise training.
Blog: research.google/blog/sensorlm-learning-the-language-of-wearable-sensors



BigSmall (WACV '24)

June '22 - Jan. '24

- Efficient multitask physiological sensing for spatially and temporally disparate signals
- Simultaneous modeling of pulse, respiration, and facial action from face video
- Website: [girishvn.github.io/BigSmall](https://github.com/girishvn/BigSmall)



rPPG Toolbox (NeurIPS '23)

June '22 - Nov. '23

- Enables standardized and accessible remote photoplethysmography (rPPG) research
- 750+ stars, 190+ forks.
- Repo: github.com/ubicomplab/rPPG-Toolbox

Work Experience



UW Ubiquitous Computing Lab Research Assistant

Sept. '20 - Present

- Research focused on novel time-series ML/AI methods applied to health.



Google Student Research

June '23 - Sept. '25

- Foundation models for wearable data ([LSM-1 \[8\]](#), [LSM-2 \[UR.6\]](#), [SensorLM \[UR.5\]](#)).
- Audio models for respiratory sensing. Models can be found on [Github](#) / [HuggingFace](#).

-  **Uplinq** *Machine Learning Engineer* Apr. '22 - Nov. '22
- Startup making bookkeeping and operations data accessible for small businesses
 - Developed system to automate bookkeeping using deterministic and ML solutions
-  **OctoAI** *Machine Learning Systems Intern* July '21 - Oct. '21
- Developed system to offload ML inference workload from MCU to FPGA accelerator.
 - Pipeline to interface with OctoAI's uTVM ML accelerator runtime.
-  **CU Hearing Research Lab** *Research Assistant* Aug. '19 - Mar. '20
- Developed tools and GUIs for use in audiology clinical studies.
 - ML models to predict audio quality and perception.
-  **Qualcomm Technologies: Wifi Team** *R&D Software Intern* June '18 - Sept. '18
- Built framework to package, store, and visualize 11ax Wifi PHY simulation results
-  **CU Correll Lab** *Undergraduate Research Assistant* Oct. '16 - June '17
- UROP grant to color characterize novel force-proximity sensor used for a robotic claw
 - Assisted in integration of the sensor onto a prosthetic hook
-  **Medtronic: Surgical Devices R&D Team** *R&D Software Intern* May '16 - Aug. '16
- Device drivers for Ultrasonic Tissue Dissector

Side Projects

-  **Disease Spread in Small Population Networks** Apr. '19 - May '19
- Graph analysis to test similarity of sampled-co-presence and face-to-face networks
 - Applied to disease spread and vaccination in small population graphs (100-500 nodes)

Publications Under Review

* denotes equal contribution

[UR.5] **Insulin Resistance Prediction From Wearables and Routine Blood Biomarkers**

Nature

Metwally, Heydari, McDuff, Solot, Esmaeilpour, Faranesh, Zhou, Narayanswamy, Xu, Liu, Savage, Malhotra, Heneghan, Patel, Speed, Prieto

[UR.4] **LSM-2: Learning from Incomplete Wearable Sensor Data**

International Conference on Learning Representations (ICLR '26)

Narayanswamy*, Xu*, Ayush, Spathis, Liao, Taylor, Metwally, Heydari, Zhang, Garrison, Abdel-Ghaffar, Xu, Gu, Sunshine, Poh, Liu, Althoff, Narayanan, Kohli, Malhotra, Patel, Yang, Rehg, Liu, McDuff

[UR.3] **The Anatomy of a Personal Health Agent**

Nature

Heydari, Gu, Srinivas, Yu, Zhang, Zhang, Paruchuri, He, Palangi, Hammerquist, Metwally, Winslow, Kim, Ayush, Yang, Narayanswamy, Xu, Garrison, Lee, Vafeiadou, Graef, Galatzer-Levy, Schenck, Barakat, Perez, Shreibati, Hernandez, Faranesh, Prieto, Heneghan, Liu, Zhan, Malhotra, Patel, Althoff, Liu, McDuff, Xu

[UR.2] **Estimating Blood Pressure with a Camera: An Exploratory Study of Ambulatory Patients with Cardiovascular Disease**

NPJ Digital Medicine

Curran, Ma, Liu, McDuff, Narayanswamy, Stergiou, Patel, Yang

[UR.1] **CapApp: Smartphone-Based Capillary Refill Time Assessment**

ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT)

Park, Wu, Sarda, Narayanswamy, Strutt, Thompson, Patel, Mariakakis

Publications

* denotes equal contribution

[11] **SensorLM: Learning the Language of Wearable Sensors**

Advances in Neural Information Processing Systems (NeurIPS '25)

Zhang, Ayush, Qiao, Heydari, Narayanswamy, Xu, Metwally, Xu, Garrison, Xu, Althoff, Liu, Kohli, Zhan, Malhotra, Patel, Mascolo, Liu, McDuff, Yang

[10] **RADAR: Benchmarking Language Models on Imperfect Tabular Data**

Advances in Neural Information Processing Systems (NeurIPS '25)

Gu, Zhang, Lin, Zhang, Paruchuri, Yu, Kazemi, Ayush, Heydari, Xu, Narayanswamy, Liu, Poh, Yang, Malhotra, Patel, Palangi, Xu, McDuff, Althoff, Liu

[9] **Autonomous collection of voiding events for sound uroflowmetries with machine learning**

Biomedical Signal Processing and Control '25

Arjona, Hernandez, Narayanswamy, Bahillo, Patel

[8] **Scaling Wearable Foundation Models**

International Conference on Learning Representations (ICLR '25)

Narayanswamy*, Liu*, Ayush, Yang, Xu, Liao, Garrison, Tailor, Sunshine, Liu, Althoff, Narayanan, Kohli, Zhan, Malhotra, Patel, Abdel-Ghaffar, McDuff

[7] **Exploring and characterizing large language models for embedded system development and debugging**

ACM Conference on Human Factors in Computing Systems Late-Breaking Work (CHI LBW '24)

Englhardt, Li, Nissanka, Zhang, Narayanswamy, Breda, Liu, Patel, Iyer

[6] **BigSmall: Efficient Multi-Task Learning for Disparate Spatial and Temporal Physiological Measurements**

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV '24)

Narayanswamy, Liu, Yang, Ma, Liu, McDuff, Patel

[5] **rPPG-Toolbox: Deep Remote PPG Toolbox**

Advances in Neural Information Processing Systems (NeurIPS '23)

Liu, Narayanswamy*, Paruchuri*, Zhang, Tang, Zhang, Sengupta, Patel, Wang, McDuff

[4] **Camera-Based Remote Photoplethysmography for Blood Pressure Measurement: Current Evidence, Clinical Perspectives, and Future Applications**

Connected Health and Telemedicine '23

Curran, McDuff, Liu, Narayanswamy, Ma, Patel, Yang

[3] **Camera-based remote photoplethysmography to predict blood pressure in clinic patients with cardiovascular disease**

Journal of Hypertension '23

Curran, Liu, McDuff, Ma, Narayanswamy, Patel, Yang

[2] **CapApp: Smartphone Based Capillary Refill Index Assessment in Healthy Children**

Frontiers in Biomedical Devices '23

Strutt, Narayanswamy, Park, Sarda, Wu, Thompson, Harvey, Hedstrom, Kodet, Patel, Mariakakis

[1] **Automatic Classification of Audio Uroflowmetry with a Smartwatch**

International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC '22)

Narayanswamy, Arjona, Diez, Bahillo, Patel

Teaching and Mentorship

 UbiComp Lab <i>Research Mentor</i>	June '22 - Present
• Kanav Arora (Efficient models for wearable data) • Liam Bakar (AI assisted food tracking) • Dilini Nissanka (AI assisted food tracking) • Nancy (Yujia) Liu (now PhD student at UCSD) • Jack (Chengqian) Ma (now ML researcher at SenseTime)	
W Graduate Application Support Program <i>Mentor</i>	Jun. '22 - Present
• Providing application feedback to applicants from under represented backgrounds	
W GIX: Access Computing Summer Program <i>Research Mentor</i>	June '22 - Sept. '22
• Program coordinated by University of Washington / Tsinghua University / Microsoft • Mentored MS student on ML methods for domain adaptation of wearable data	
W Signal Processing I <i>Teaching Assistant</i>	Sept. '21 - Dec. '21
• Audio Processing, Computer Vision, Filter Design	
W Ubiquitous Computing Seminar <i>Instructor</i>	Jan. '21 - Mar. '21
• Developed seminar format, organized participation and speakers	
W Computer Vision - Classical & Deep Methods <i>Teaching Assistant</i>	Sept. '20 - Dec. '20
• OpenCV, PyTorch, Image Processing, Deep Learning	
 Applications of Embedded Systems <i>Teaching Assistant</i>	Aug. '16 - Dec. '16
• Embedded C, TI MSP432 ARM Cortex M4 MCU, software / hardware Debug	

Organizing and Service

 Time-Series For Health: NeurIPS '25 <i>Workshop Organizer</i>	June '25 - Dec. '25
• Organized reviewing, invited speakers, and workshop event. • 120+ workshop paper submissions, 230+ reviewers • timeseries4health.github.io	
 Reviewing <i>Workshop Organizer</i>	June '25 - Dec. '25
• NeurIPS, CHI, IMWUT, Journal of Biomedical and Health Informatics, Nature Scientific Reports, ACM COMPASS '22 Posters TCP	

Selected Skills

Domain Expertise AI / ML, Self-Supervised Learning, Generative Pre-Training, Foundation Models, Signal Processing, Computer Vision, Time-Series Models, Audio Models, Health Sensing, Embedded Systems

Languages Python, C / C++, MATLAB

Frameworks / Libraries / Version Control PyTorch, Jax, Tensorflow, Scikit-Learn, Pandas, Numpy, Git

Platforms Linux / Unix, ARM-Based MCUs, Android

EE Bench-top Skills PCB Design, SMT / Through-Hole Soldering, Oscilloscope, Function Generator, Logic Analyzer, Multimeter, Hardware Debugging

Mathematics Calc. I / II / III, Differential Eqs., Linear Algebra, Discrete Math, Probability, Matrix Methods

Other



Awards and Honors

- Engineering Honors Program (CU Boulder)
- Latin (Magna Cum Laude) and Departmental Honors
- ECE Outstanding Colleague Award (CU Boulder)
- Dean's List (CU Boulder): All semesters (Fall '14 - Spring '19)
- Ugrad Scholarships: Intel Merit, BOLD, Engineering Differential, Sewall Esteemed, UROP Individual Grant



Extracurricular Activities

- Boulder Lotus ('21): One of the nation's top 15 club ultimate frisbee teams
- University of Colorado Mamabird ('14 - '19): One of the nation's top 5 collegiate ultimate frisbee teams
- Fairview HS Ultimate Frisbee Coach ('19 - '20)
- Shotokan Karate: First degree black belt and instructor at International Martial Arts Association
- Interests: Photography (ig: [gvn_photos](#)), Drawing, Hiking, Camping, Skiing